

What UAT confirms

In a test environment you confirm the vendor's randomization and drug-supply build does **exactly** what your URS and specs say, before any real participant is randomized. You operate the staging IRT in a browser, capture dated screenshot evidence, and transcribe results into your workbook so the record is attributable and stands alone for an auditor.

The go-live gate (conjunctive)

Sign-off is READY only when **all three** hold:

- Every **Critical** test passes
- **0** open Critical / Major defects
- **100%** traceability (every requirement has a test)

Snapshot now: **NOT ready - blocked**.

Severity, not count, decides. The gate packages evidence; a named human signs.

Live snapshot

Test scripts	29
Passed	22
Failed	2
Open defects	3
Requirements traced	97%

Defect severity drives the gate

Critical	Could break the blind, mis-randomize, mis-dose, or corrupt integrity	Blocks
Major	Significant functional error in a key workflow (resupply, wrong kit, missing alert)	Blocks til fixed
Minor	Cosmetic / non-functional (label typo, missing site number)	Fix or waive

Where it runs: the systems (the workbook is the paper trail)

Staging IRT (vendor, test list) → **Operate in a browser** → **Dated screenshots + exports** → **Excel workbook** (status-defect-eTMF (signing gate) package)

Failures also go to the **vendor defect tracker**; the vendor promotes a fixed build back to staging for retest. The workbook never connects to the IRT, you transcribe, on purpose.

Anatomy of an executable test (RAND-02)

A good test has a **specific, verifiable** expected result, not "works correctly".

- **Precondition:** test list loaded; the <70 kg stratum, block 1, has an unused slot.
- **Steps:** register 101-001 at 68.0 kg → randomize → record kit + confirmation → open the allocation report as the unblinded statistician.
- **Expected:** allocated to the next unused <70 kg / block-1 entry; arm, sequence, kit match the list; the investigator sees a **kit ID only**, not the arm.
- **Verify:** compare cell-for-cell across every stratum and both arms, not one happy-path participant.

The 8-step workflow

1. Derive test cases from URS
2. Load the approved test list + specs (positive · negative · into staging IRT boundary)
3. Execute each case in the browser; be systematic
4. Record result + dated screenshot evidence
5. Log every failure as a defect with a severity
6. Vendor fixes; retest the new build
7. Traceability: every requirement maps to a passing test
8. Evaluate the gate; a named human signs it into the eTMF

Run the interactive tool

```
install.packages c('shiny', 'bslib', 'DT', 'ggplot2')
shiny::runApp('uat_irt/shiny')
```

Select a test, mark it Pass/Fail, close a defect, and the

